

Information-Theoretic SCX

——rate-distortion

Information-Theoretic SCX: Rate-Distortion Theory of
Multi-Expert Information Fusion

SCX

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Abstract

SCXState-Conditioned eXpertise M howeverSCXtheoremTheorem 1–4Core communicationupper boundencodingcompression-channelencoding theorem——rate-distortionSlepian-WolfencodingShannontheorem—— SCX

rate-distortionSCXrate-distortionVersion $Q(\mathbf{V})\mathbf{V} \in \{0, 1\}^M$ communicationStatus R communicationSCXlower bound $P_e \geq (H(Y) - R - \log 2) / \log |\mathcal{Y}|$ -**communication-** Bottleneck-compression

Slepian-Wolfencoding M source ProofSlepian-WolftheoremSCXVersion M compression $\sum_{m \in \mathcal{T}} R_m \geq H(\mathbf{V}_{\mathcal{T}} | \mathbf{V}_{\mathcal{T}^c})$ all $\mathcal{T}H(\mathbf{V})M \cdot H_2(p_{\text{clean}})$ encoding—— SCX[**Honest Strike**]

ShannontheoremSCXProofSCXthere exists iencodingcompressionScoreii channelencodingif and only ifsatisfying otherwise-channelencoding

Theorem 2Theorem 2Fanoawaiting rate-distortionBottleneck $\delta(R) = \inf_{P_{U|X}: I(X;U) \leq R} I(Y;U)$ Proof R awaitingTheorem 2upper bound $F_{1,\text{SCX}} \leq F_{1,\text{base}} + C_F \cdot r(\delta(0))$ wherer $(\cdot)XY$ rate-distortion

Honest StrikeRigorous Prooftheoremassumption SCX

- rate-distortionCompactnessi.i.d.

Assumption 0.1. ——*Status*there exists

Slepian-Wolfencodingfeasibilityencodingthere exists common randomness——SCX

theoremSCXM $\rightarrow \infty$, $n \rightarrow \infty$

Theorem 2although $\delta(R)$ Theorem 2——there exists-

KeywordsSCXrate-distortionSlepian-WolfencodingShannontheoremBottleneck Mutual InformationFanoawaiting

1 SCX

1.1 SCX

SCX MN Status Coretheorem[11]

- **Theorem 1**Consistency $F_1 \geq 1 - \frac{1}{\eta} \sum_s \rho_s \exp(-2M\Delta_s^2)$
- **Theorem 2** $F_{1,\text{SCX}} \leq F_{1,\text{base}} + C_F \sqrt{\delta/2}$
- **Theorem 3**-there existsawaiting
- **Theorem 4**MinimaxChernoff-Steinlower bound

howevertheoremthere exists Theorem 1Chernoff-HoeffdingawaitingTheorem 2Fanoawaiting Theorem 4Chernoff-Steinlemma—— Core——rate-distortionsourceencodingtheoremBottleneck—— SCX

CoreSCX M **sourcedestination** communicationencoding **channel**Perspectivetheoremassumption

1.2 SCX

Definition 1.1 (SCX). *SCX* define

$$\boxed{\mathcal{T}_{SCX} = (\mathcal{S}, \mathcal{E}, \mathcal{C}, \mathcal{A}, \mathcal{D})} \quad (1)$$

where

- $\mathcal{S} \text{Status} NY_s \in \mathcal{Y}$
- $\mathcal{E} = \{E_1, \dots, E_M\}$ each $E_m \text{Status} V_m(s) = \mathbf{1}[E_m(\phi(s)) \neq y_s] \in \{0, 1\}$
- $\mathcal{C} \text{communication channel} \mathbf{V}(s) = (V_1(s), \dots, V_M(s))$ compression $Q(\mathbf{V}(s)) \in \{1, \dots, 2^R\}$ satisfying $R = \frac{1}{N} \sum_s \log |Q(\mathbf{V}(s))|$
- $\mathcal{A} \text{compression} \hat{Y}_s = \mathcal{A}(Q(\mathbf{V}(s)))$
- $\mathcal{D} d(\hat{Y}_s, Y_s) = \mathbf{1}[\hat{Y}_s \neq Y_s]$

SCX-channel encoding M source $\rightarrow$ encoding compression $\rightarrow$ channel communication $\rightarrow$ rate-distortion Slepian-Wolf theorem $\rightarrow$ SCX

Definition 1.2. *Status* $y_s M \mathbf{V} \in \{0, 1\}^M$

$$P(\mathbf{V} = \mathbf{v} \mid Y = y) = \prod_{m=1}^M p_y^{v_m} (1 - p_y)^{1-v_m} \quad (\text{assumption}) \quad (2)$$

where $p_{\text{clean},s} = P(V_m = 1 \mid Y = \text{clean})$ $p_{\text{noisy},s} = P(V_m = 1 \mid Y = \text{noisy})$
assumption define 1.1 SCX *assumption* §3§5

[Honest Strike] Honest Strike SCX Core *assumption* Status satisfying asymptotic $\rightarrow$ iias-
 sump tion communication $\rightarrow$ Bottleneck asymptotic

2 rate-distortion communication SCX lower bound

2.1 rate-distortion SCX

rate-distortion Rate-Distortion Theory source compression Core [4] source $X \sim P_X d(x, \hat{x})$ rate-distortion define

$$R(D) = \min_{P_{\hat{X}|X}: \mathbb{E}[d(X, \hat{X})] \leq D} I(X; \hat{X}) \quad (3)$$

SCX “source” $M \mathbf{V} \in \{0, 1\}^M$ “” $P_e = \mathbb{E}[d(\hat{Y}, Y)] = P(\hat{Y} \neq Y)$ “” Status communication R
 Core communication R SCX lower bound

2.2 theorem rate-distortion lower bound

Theorem 2.1 (SCX rate-distortion lower bound $\rightarrow$). M Status $\mathbf{V} \in \{0, 1\}^M$ Y $\mathbf{V}$ Version $Q(\mathbf{V})$ satisfying $I(\mathbf{V}; Q(\mathbf{V})) \leq R$ Status satisfying

$$\boxed{P_e \geq \frac{H(Y) - R - \log 2}{\log |\mathcal{Y}|}} \quad (4)$$

awaiting ε communication

$$\boxed{R_{\min}(\varepsilon) \geq H(Y) - \log 2 - \varepsilon \log |\mathcal{Y}|} \quad (5)$$

Proof. **1Fanoawaiting**Fanoawaiting[4]

$$H(Y | Q(\mathbf{V})) \leq H_2(P_e) + P_e \cdot \log(|\mathcal{Y}| - 1) \leq \log 2 + P_e \log |\mathcal{Y}| \quad (6)$$

where $H_2(P_e) \leq \log 2$ Entropyupper bound

2Entropyawaiting

$$H(Y | Q(\mathbf{V})) = H(Y) - I(Y; Q(\mathbf{V})) \quad (7)$$

$$\geq H(Y) - I(\mathbf{V}; Q(\mathbf{V})) \quad \text{DPI: } Y \rightarrow \mathbf{V} \rightarrow Q(\mathbf{V}) \quad (8)$$

$$\geq H(Y) - R \quad (9)$$

3lower bound

$$H(Y) - R \leq H(Y | Q(\mathbf{V})) \leq \log 2 + P_e \log |\mathcal{Y}| \quad (10)$$

$$\implies P_e \geq \frac{H(Y) - R - \log 2}{\log |\mathcal{Y}|} \quad (11)$$

4Compactnesslower boundasymptotic i $Q(\mathbf{V})$ Y sufficient statistic i.e. $Y \perp\!\!\!\perp \mathbf{V} | Q(\mathbf{V})$ ii Fanoawait-
ingawaitingall iiiawaitingi.e. $I(\mathbf{V}; Q(\mathbf{V})) = R$ $M \rightarrow \infty$ awaiting $\square$ $\square$

[Rigorous Proof] ProofStatus:1–3Fanoawaiting+DPI+Entropy 4Compactnessasymptotic——

Remark 2.2 (Theorem 1——Chernoff vs. rate-distortion). Theorem 1 F_1 lower boundChernoff-
Hoeffdingtheorem2.1 P_e lower boundrate-distortion

• Theorem 1 M —— $M\Delta_s > 0$ F_1 1“”

• theorem2.1**communication**—— M communication R lower bound“”

Theorem 1assumptioncommunicationall theorem2.1assumptionchannel rate-distortion

2.3 rate-distortion

Corollary 2.3 (-communication-). *SCX*there exists

$$\underbrace{R}_{\text{communication}} \cdot \underbrace{N}_{\text{Status}} \geq H(\mathbf{V}_1, \dots, \mathbf{V}_M) - \underbrace{N \cdot H_2(P_e)} \quad (12)$$

$MH(\mathbf{V})$ encodingsimultaneously Δ_s Growth $P_e R$ there exists M^*

Proof. rate-distortiondefinetheorem2.1 $H(\mathbf{V})$ M Growth Growth $M P_e M \exp(-2M\Delta_s^2)$ Theorem 1
 M^* $\square$ $\square$

[Heuristic] ProofStatus:Heuristic $M^* H(\mathbf{V}) P_e(M)$ $MH(\mathbf{V})$ ——SCX

2.4 BottleneckSCXcompression

rate-distortionlower boundcompressionBottleneck Information Bottleneck, IB[5]

Definition 2.4 (SCXBottleneck). *SCXBottleneck*

$$\min_{P_{Q|\mathbf{V}}} I(\mathbf{V}; Q) - \beta \cdot I(Y; Q) \quad (13)$$

where Q compression $\beta > 0$ compression $\beta \rightarrow 0$ compression $Q\mathbf{V}$ $\beta \rightarrow \infty$ Q ally

Proposition 2.5 (SCX-IBcompression). *source* $\{0, 1\}^M$ SCX-IBsatisfying

$$P_{Q|V}(q | \mathbf{v}) = \frac{P_Q(q)}{Z(\mathbf{v}, \beta)} \exp(-\beta \cdot D_{\text{KL}}(P_{Y|\mathbf{V}=\mathbf{v}} \| P_{Y|Q=q})) \quad (14)$$

where $Z(\mathbf{v}, \beta) = \sum_q P_{Q|V}(q | \mathbf{v})$ and $P_{Y|Q} = \sum_{\mathbf{v}} P_{Y|\mathbf{V}=\mathbf{v}} P_{Q|V}(q | \mathbf{v})$

Proof. Tishby [5] Core SCX source source $X \mathbf{V} Y$ IB convexity $P_{Q|V} P_{Y|Q}$ $\square$ $\square$

[**Rigorous Proof**] **Proof Status:** IBconvergence SCX $P_{Y|\mathbf{V}}$ IBcompression

[**Honest Strike**] **Honest Strike** IBfeasibility although IBSCX 2^M — $M = 10$ i.e. 1024 Status $M = 20$ i.e. for awaiting IBIB [6]

2.5 Masymptoticrate-distortion

Theorem 2.6 (asymptotic — Strassen). *for* MN Status communication N RD *there exists* ϵ *if and only if* there exists $P_{Q|\mathbf{V}}$ such that

$$\boxed{\frac{1}{N} \sum_{s=1}^N I(\mathbf{V}_s; Q_s) \leq R \quad \frac{1}{N} \sum_{s=1}^N \mathbb{E}[d(\hat{Y}_s, Y_s)] \leq D} \quad (15)$$

N there exists **channel** channel dispersion $O(1/\sqrt{N})$ such that asymptotic $\sqrt{V(D)/N} \cdot Q^{-1}(\epsilon)$ where $V(D)$ rate-distortion Q^{-1}

Proof. Kostina & Verdú [7] asymptotic rate-distortion SCX asymptotic Berry-Esseen $\square$ $\square$

[**Rigorous Proof**] **Proof Status:** channel encoding Strassen 1962 Hayashi 2006 $V(D)$ source —

SCX all $p_{\text{clean},s} p_{\text{noisy},s}$

for SCX $N \approx 10^3$ – 10^6 Status asymptotic 10%–30% communication

3 Slepian-Wolf encoding M compression

3.1 encodingsourceencoding

SCX communication M — each encoding encoding communication source encoding Distributed Source Coding

Definition 3.1 (SCX encoding). M encoding $V_m \in \{0, 1\}$ encoding m $C_m(V_m) \in \{0, 1\}^{nR_m}$ all $(C_1, \dots, C_M)$ $\hat{\mathbf{V}} = (\hat{V}_1, \dots, \hat{V}_M)$ $\lim_{n \rightarrow \infty} P(\hat{\mathbf{V}} \neq \mathbf{V}) = 0$ n Status 0

3.2 Slepian-Wolf theorem SCX Version

Theorem 3.2 (SCX Slepian-Wolf encoding theorem — compression). $M \mathbf{V} = (V_1, \dots, V_M) P_{\mathbf{V}}$ there exists encoding any $\mathbf{V}$ if and only if $(R_1, \dots, R_M)$ satisfying

$$\boxed{\sum_{m \in \mathcal{T}} R_m \geq H(\mathbf{V}_{\mathcal{T}} | \mathbf{V}_{\mathcal{T}^c}), \quad \forall \mathcal{T} \subseteq \{1, \dots, M\}} \quad (16)$$

where $\mathbf{V}_{\mathcal{T}} = \{V_m : m \in \mathcal{T}\}$ $\mathbf{V}_{\mathcal{T}^c} = \{V_m : m \notin \mathcal{T}\}$ in **particular** satisfying

$$\boxed{\sum_{m=1}^M R_m \geq H(\mathbf{V})} \quad (17)$$

$H(\mathbf{V}) = \sum_m H(V_m) = M \cdot H_2(p)$ Slepian-Wolf encoding $\text{Cov}(V_i, V_j) \rightarrow \text{Var}(V_i)$ $H(\mathbf{V}) \ll \sum_m H(V_m)$ — encoding

Proof. 1 random binning each encoding $m \mathbf{v}_m^n 2^{nR_m}$ bin all lemma 1

2 **theorem** $\mathcal{T}$ awaiting Entropy compression $\geq 1/2$ — Fano awaiting lower bound

3 V_m $H(\mathbf{V}) = \sum_m H(V_m) = \sum_m H_2(p_{\text{clean},s})$ all Slepian-Wolf $R_m \geq H(V_m)$ — encoding $\square$
 $\square$

[**Rigorous Proof**] **Proof Status:** theorem 3.2 Slepian-Wolf theorem [2] SCX source SCX Slepian-Wolf — because source $\{0, 1\}^M 2^M$

3.3 —

Theorem 3.3 (— Slepian-Wolf SCX). $\rho_{ij} = (V_i, V_j)_{ij}$ Slepian-Wolf compression for encoding

$$G_{SW}(\rho) = \frac{\sum_{m=1}^M H(V_m)}{H(\mathbf{V})} - 1 \geq 0 \quad (18)$$

$G_{SW} \rho_{ij} \rightarrow 1 H(\mathbf{V}) \rightarrow H(V_1)$ 1 encoding all M Theorem $1 \Delta_s M_{\text{eff}} 0$ — **compression SCX**

Proof. 1 **Entropy** Entropy $H(\mathbf{V}) = \sum_{m=1}^M H(V_m | V_1, \dots, V_{m-1}) \leq \sum_{m=1}^M H(V_m) \rightarrow$ Entropy $H(V_m | V_{<m}) \rightarrow H(\mathbf{V}) \rightarrow G_{SW}$

2 theorem 2.5.1 M_{eff} $M_{\text{eff}} = M / (1 + (M-1)\bar{\rho})$ $\bar{\rho} \rightarrow 1 M_{\text{eff}} \rightarrow 1$ — $\Delta_s M$ Decays

3 $G_{SW} M_{\text{eff}} \bar{\rho}$ $G_{SW} \uparrow \iff \bar{\rho} \uparrow \iff M_{\text{eff}} \downarrow$ “” **communication reliability** $\square$ $\square$

[**Honest Strike**] **Honest Strike** Slepian-Wolf SCX Slepian-Wolf encoding —

(1) encoding $P(V_1, \dots, V_M)$ — SCX $p_{\text{clean},s} p_{\text{noisy},s}$ encoding $\rightarrow$ Fails

(2) all encoding — communication

(3) Slepian-Wolf asymptotic optimality $n \rightarrow \infty$ “encoding” — SCX $n = 1$ awaiting for $n = 1$ Slepian-Wolf encoding

Corollary 3.4 (encoding Slepian-Wolf). $n = 1$ Slepian-Wolf encoding **encoding**

$$R_m \geq H(V_m) = H_2(p_{\text{clean},s}) H_2(p_{\text{noisy},s}) \quad (19)$$

encoding $n \gg 1$ batch auditing

[**Open Problem**] **Open Problem 1** Slepian-Wolf there exists SCX zero-delay, $n = 1$ encoding i.e. — Slepian-Wolf theorem — SCX $[0, 1]$

3.4 assumption Slepian-Wolf

satisfying SCX Slepian-Wolf $R_m \geq H_2(p_{\text{clean},s})$ there exists because $H(V_i | V_j) \leq H(V_i)$ i.e.

SCX “communication” $\rightarrow$ Slepian-Wolf $\rightarrow H(V_m) \rightarrow$ communication $M \cdot H_2(p)$ SCX Conclusion — reliability communication

4 Shannon theorem SCX encoding channel encoding

4.1 theorem SCX

Shannon **theorem** Separation Theorem [1] communications source encoding compression Channel Capacity channel encoding channel optimality theorems such that communication — compression SCX there exists

• **SCX “encoding”** M compression **Score** $Q(s)$ — “compression” 2^M Status compression Q

• **SCX “channel encoding”** / Verdict — “”

SCX encoding channel encoding

4.2 SCXtheorem

Theorem 4.1 (SCXtheorem—). *There exists*

(i) **encoding** $\mathbf{V}$ compressionsufficient statistic $T(\mathbf{V}) = \sum_{m=1}^M w_m V_m$ where $w_m^* = \log \frac{p_{\text{noisy},m}(1-p_{\text{clean},m})}{p_{\text{clean},m}(1-p_{\text{noisy},m})}$

(ii) **channelencoding** $T(\mathbf{V}) \rightarrow Y$ Verdict

F_1 Score

Proof. 1sufficient statisticassumption

$$\Lambda(\mathbf{V}) = \sum_{m=1}^M \log \frac{P(V_m | Y = \text{noisy})}{P(V_m | Y = \text{clean})} = \sum_{m=1}^M V_m \cdot w_m^* + \text{const} \quad (20)$$

Fisher-Neymantheorem $T(\mathbf{V}) = \sum_m w_m^* V_m$ sufficient statistic

2optimality sufficient statistic $Y \perp\!\!\!\perp \mathbf{V} | T(\mathbf{V})$ therefore $\mathbf{V} T(\mathbf{V})$ sourceencoding $\mathbf{V} \rightarrow T(\mathbf{V}) Y$ channelencoding $T(\mathbf{V})$

3assumption $T(\mathbf{V}) = \sum_m w_m V_m$ sufficient statistic— Y -channelencoding $\square \quad \square$

[**Rigorous Proof**] **ProofStatus:** 1–2sufficient statistic 3awaiting Conclusion if and only if Markov $Y \rightarrow T(\mathbf{V}) \rightarrow \mathbf{V} T(\mathbf{V})$

Theorem 4.2 (theorem—encoding). *there exists* $\text{Cov}(V_i, V_j) \neq 0$ -channelencoding

$$F_1^{\text{joint}} \geq F_1^{\text{separate}} + \underbrace{\frac{I(Y; \mathbf{V}) - I(Y; T(\mathbf{V}))}{\log |\mathcal{Y}|}}_{\text{lower bound}} - O\left(\frac{1}{\sqrt{M}}\right) \quad (21)$$

Y

Proof. awaiting Fano awaiting $T(\mathbf{V})$ sufficient statistic $\mathbf{V}$ Mutual Information $I(Y; \mathbf{V}) - I(Y; T(\mathbf{V}))$ lower bound Cover & Thomas[4] Fano DPI $\square \quad \square$

[**Rigorous Proof**] **ProofStatus:** $I(Y; \mathbf{V}) - I(Y; T(\mathbf{V}))$ $\mathbf{V}$ —

4.3 theorem vs.

Corollary 4.3 (SCX). (1) $\rightarrow$ Verdict

(2) *there exists*

(3) lower bound — Overfitting

[**Honest Strike**] **Honest Strike** theorem asymptotic Shannon theorem asymptotic $n \rightarrow \infty$ SCX $n = 1$ due to insufficient statistic $T(\mathbf{V})$ sufficiency asymptotic— $MT(\mathbf{V}) Y$ iiassumption Mp -hacking

asymptotic theorem [8] SCX channelencoding

5 Theorem 2 Fano rate-distortion Bottleneck

5.1 Theorem 2

Theorem 2 [11]

$$F_{1,SCX} \leq F_{1,base} + C_F \cdot \sqrt{\frac{\delta}{2}} \quad (22)$$

where $\delta = I(Y; X_{ideal}) - I(Y; X)$ Mutual Information C_F Theorem 2' define $3.1 \varepsilon_{PE}$ encoding $\varepsilon_{PE} = I(Y; P | X) - I(Y; PE(P) | X)$ upper bound

$$F_{1,SCX}^{SITUS} \leq F_{1,base} + C_F \cdot \sqrt{\frac{\delta + 2\varepsilon_{PE}/C_F^2}{2}} \quad (23)$$

however there exists

- (1) Fano awaiting P_e lower bound $\eta < 0.5$ — lower bound vacuous because $H_2(\eta) < \log 2(H_2(\eta) - \log 2) < 0$
- (2) **rate-distortion** δ — i.e. rate-distortion $R(D)$ Bottleneck
- (3) $\delta = I(Y; X_{ideal}) - I(Y; X)$ $P_{Y|X}$ δ

5.2 $\delta(R)$ — rate-distortion

Definition 5.1 ($\delta(R)$). $X X_{ideal} Y$ define

$$\boxed{\delta(R) = I(Y; X_{ideal}) - \max_{P_{U|X}: I(X; U) \leq R} I(Y; U)} \quad (24)$$

where $R \geq 0$ encoding X $\delta(0) = I(Y; X_{ideal}) - I(Y; X) = \delta_{original}$ define $\delta(\infty) = 0$ $R U X$ $\delta(R)$ R convexity $P_{X,Y}$

Theorem 5.2 (Theorem 2 rate-distortion —). $SCX X$ compression U satisfying $I(X; U) \leq R$ F_1 Score upper bound

$$\boxed{F_{1,SCX}(R) \leq F_{1,base} + C_F \cdot r(\delta(R))} \quad (25)$$

where $r(\delta) = \sqrt{\delta/(2 \log |\mathcal{Y}|)}$ rate-distortion- in particular

- (i) $R \rightarrow 0$ $\delta(R) \rightarrow \delta(0) = \delta$ upper bound Theorem 2
- (ii) $R \rightarrow \infty$ $\delta(R) \rightarrow 0$ upper bound $F_{1,base}$
- (iii) for R upper bound Theorem 2 $R = 0$ $R = \infty$

Proof. **1 rate-distortion Fano** compression $U \leq R$

$$P_e(R) \geq \frac{H(Y) - I(Y; U) - \log 2}{\log |\mathcal{Y}|} \quad (26)$$

2

$$I(Y; U) = I(Y; X_{ideal}) - [I(Y; X_{ideal}) - I(Y; U)] \quad (27)$$

$$\leq I(Y; X_{ideal}) - \delta(R) \quad (28)$$

$$= H(Y) - H(Y | X_{ideal}) - \delta(R) \quad (29)$$

3 F_1 $21 H(Y | X_{ideal}) = 0$ Y define $P_e(R) \geq \delta(R)/\log |\mathcal{Y}| - O(1/\log |\mathcal{Y}|)$ $F_1 P_e$ Theorem 2 C_F upper bound

4 convexity monotonicity $\delta(R)$ Bottleneck $\max_{P_{U|X}: I(X; U) \leq R} I(Y; U)$ R Tishby awaiting [5] therefore $r(\delta(R)) R$ bound $R = 0$ $R = \infty$ $\square$ $\square$

[Rigorous Proof] Proof Status: 1–3 awaiting awaiting 4 $\delta(R)$ $P_{X,Y}$ any $P_{X,Y}$ $\delta(R)$ **IB for-source** lower bound for/source

5.3 BottleneckPerspective

Theorem 2BottleneckInformation Plane

Definition 5.3 (SCX). *SCX* $I(X; U)$ compression $I(Y; U)$ define Bottleneck $I(Y; U)^*(R) = \max_{P_{U|X}: I(X; U) \leq R} I(Y; U)$ $I(Y; U) = H(Y)I(X; U)$ any

Proposition 5.4. *for* X_{ideal} IB

$$\Delta_{IB}(R) = I_{ideal}(R) - I_X(R) \quad (30)$$

where $I_{ideal}(R)I_X(R)$ Bottleneck $\Delta_{IB}(R) > 0$ all R awaiting — Theorem 2

[Heuristic] ProofStatus:Bottleneckfor $X \in \mathbb{R}^{768}$ BERTIB Alemiawaiting[6]Consistency convergenceSCX

5.4 ε_{PE} rate-distortion

ε_{PE} encodingrate-distortion

Definition 5.5 (rate-distortionencoding).

$$\varepsilon_{PE}(R) = I(Y; P | X) - \max_{P_{U|PE(P)}: I(PE(P); U) \leq R} I(Y; U | X) \quad (31)$$

$\varepsilon_{PE}(0) = I(Y; P | X)$ encoding $\varepsilon_{PE}(\infty) = \varepsilon_{PE}$ define 3.1 define

Corollary 5.6 (rate-distortionTheorem 2'). *rate-distortionTheorem 2'*

$$F_{1, SCX}^{SITUS}(R_X, R_P) \leq F_{1, base} + C_F \cdot r \left(\delta(R_X) + \frac{2\varepsilon_{PE}(R_P)}{C_F^2} \right) \quad (32)$$

where $R_X R_P$ Status X encoding $PE(P)$ $R_X R_P$ -**Pareto**

Proof. theorem 5.2 define 5.5 $XPE(P)$ Bottleneckupper boundMutual Information Mutual Information $\leq$ $\square$

[Rigorous Proof] ProofStatus: $U_X \perp\!\!\!\perp U_P | (X, PE(P))$ **assumption** otherwise $U = (U_X, U_P)$ compression $R_X R_P$ — Bottleneck

5.5 Theorem 3

rate-distortionTheorem 3-indistinguishabilityPerspective

Proposition 5.7 (Theorem 3rate-distortion). *Theorem 3indistinguishabilityawaiting* $W_A W_B$ R compression Q sat

$$\max_{W \in \{W_A, W_B\}} P_W(error | Q) \geq \frac{1}{2} - \sqrt{\frac{1}{2} D_{KL}(P_{Q|W_A} \| P_{Q|W_B})} \quad (33)$$

$P_{Q|W_A} = P_{Q|W_B}$ lower bound $\geq 1/2$ —

Proof. Le CamlemmacompressionKLT $V \geq 1$ therefore $\max \geq 1/2$ $\square$ $\square$

corollary although $PER \rightarrow \infty$ indistinguishabilityProposition 4.1 $RW_A W_B P_{Q|W}$ compression **there exists** $1/2$ PE theorem 4.1 —

6 SCX——

6.1

CoreSCX

- (1) **rate-distortion**§2SCX communicationthere existslower bound
- (2) **Slepian-Wolfencoding**§3encoding communication
- (3) **theorem**§4SCX optimality
- (4) **Theorem 2**rate-distortion§5

Mutual Informationrate-distortion SCXtheoremTheorem 1–4 **source**encoding

6.2

Table 1: SCXtheoremProof

Theorem		ProofStatus
theorem2.1	rate-distortionlower boundasymptotic	[Rigorous Proof]1–3; [Heuristic]Compactness4
theorem2.6	asymptoticrate-distortion	[Rigorous Proof]
theorem3.2	SCX Slepian-Wolftheorem	[Rigorous Proof]
theorem3.3		[Rigorous Proof]
theorem4.1	SCXtheorem	[Rigorous Proof]
theorem4.2	theorem	[Rigorous Proof]; [Heuristic]M
theorem5.2	Theorem 2rate-distortion	[Rigorous Proof]
Proposition5.4		[Heuristic]feasibility
Proposition5.7	Theorem 3rate-distortion	[Rigorous Proof]

- (1) **asymptotic vs. asymptotic** $M \rightarrow \infty N \rightarrow \infty n \rightarrow \infty$ SCXasymptotictheorem2.6 Scale
- (2) $P(\mathbf{V})P_{Y|\mathbf{V}}$ assumptionSCX ——lower bound
- (3) **Slepian-Wolf**(1)(3)§3Honest Strike
- (4) **Bottleneckfeasibility**SCXMd_XIB Open Problem

6.3 SCX

This paper revealsSCX[11, 12] Open Problem

Table 2: SCX

Status		
communication	assumption	theorem2.1lower bound
encoding	Mencoding	theorem3.2encoding
theorem	Theorem 2	theorem4.1/4.2
Theorem 2Com- pactness	Fano	theorem5.2rate-distortion

6.4 Open Problem

- (1) **Slepian-Wolf encoding** §3 SCX $n = 1$
- (2) **asymptotic SCX theorem** M theorem Version
Polyanskiy awaiting [8] channel encoding
- (3) $\delta(R)\delta(R) P_{X,Y}\delta(R)$ such that “” 95% $\delta(R) \leq \delta_0$ assumption
- (4) **Pareto** $R_X - R_P$ Status X encoding $PE(P)R_{\text{total}}$ $\delta(R_X)\varepsilon_{PE}(R_P)$ there exists Pareto
- (5) **all** Transformer
suboptimality
- (6) **SCX rate-distortion-channel encoding** Theorem 4 Minimax Chernoff-Stein lemma there exists SCX — Decays “SCX Channel Capacity”

7 Conclusion

This paper provides SCX rate-distortion Slepian-Wolf encoding Shannon theorem Bottleneck Core

Core Findings:

- (1) communication SCX lower bound $P_e \geq (H(Y) - R - \log 2) / \log |\mathcal{Y}|$ Theorem 1 Chernoff theorem 2.1
- (2) communication Slepian-Wolf compression reliability theorem 3.3
- (3) SCX “” “” assumption — theorem theorem 4.1
- (4) Theorem 2 Fano rate-distortion $R = 0$ $\delta(R)$ theorem 5.2

SCX theorem Theorem 1 Chernoff $\rightarrow$ rate-distortion theorem 2.1-lower bound Theorem 2 Fano $\rightarrow$ rate-distortion theorem 5.2 Theorem 3 indistinguishability $\rightarrow$ rate-distortion Proposition 5.7 compression Theorem 4 Minimax $\rightarrow$ Open Problem Open Problem 6 SCX Channel Capacity asymptotic assumption MN Slepian-Wolf theorem M $\delta(R)P_{X,Y}$ Fails SCX

[Honest Strike] Honest Strike The mathematical elegance of this paper SCX Bottleneck Shannon — communication Bottleneck SCX SCX —

Acknowledgments

— Shannon (1948) Slepian & Wolf (1973) Wyner & Ziv (1976) Tishby awaiting (2000) Cover & Thomas (2006) SCX Core theorem Theorem 1–4 This paper provides [theory/information.theory/DirectorySCX](#) Project, 2026 theorem

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